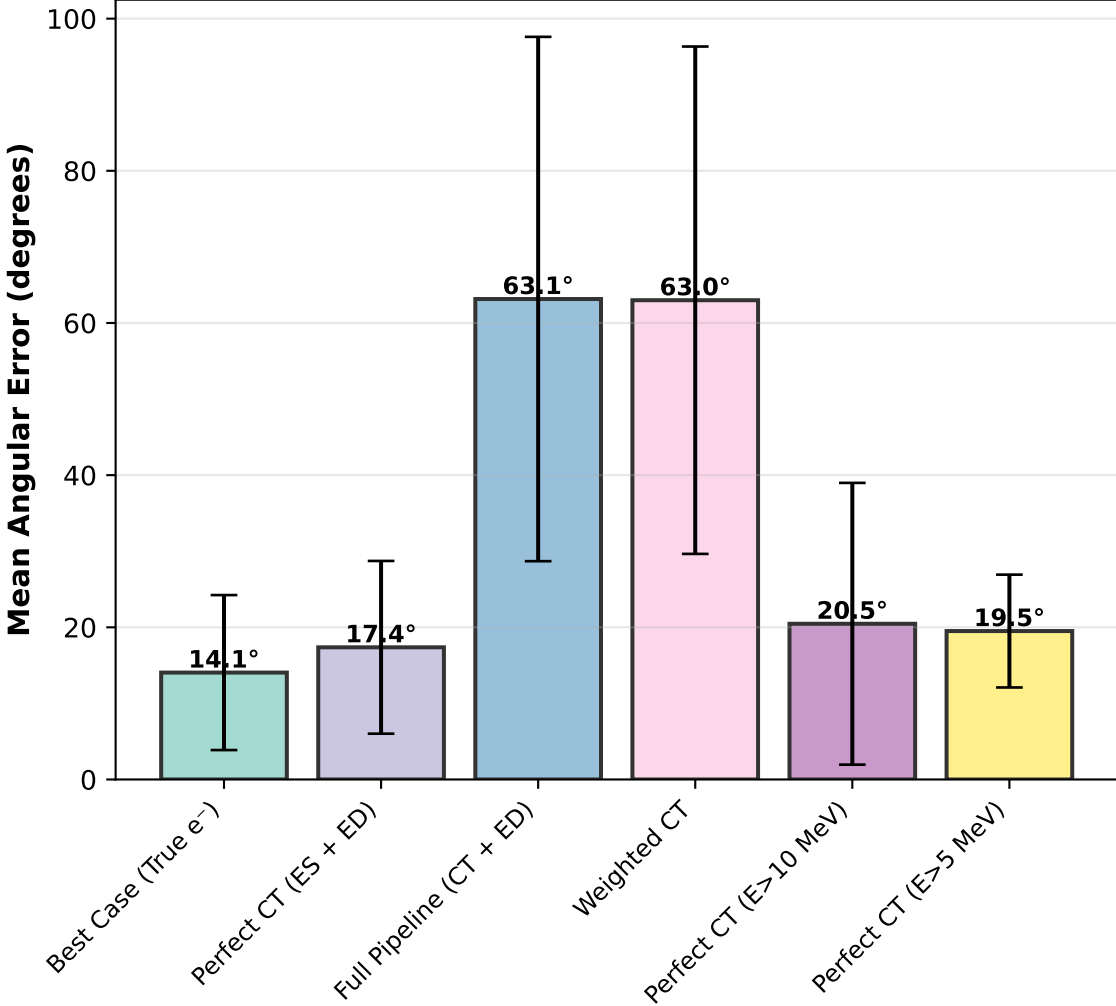


Emcee 6-Scenario Aggregate Analysis

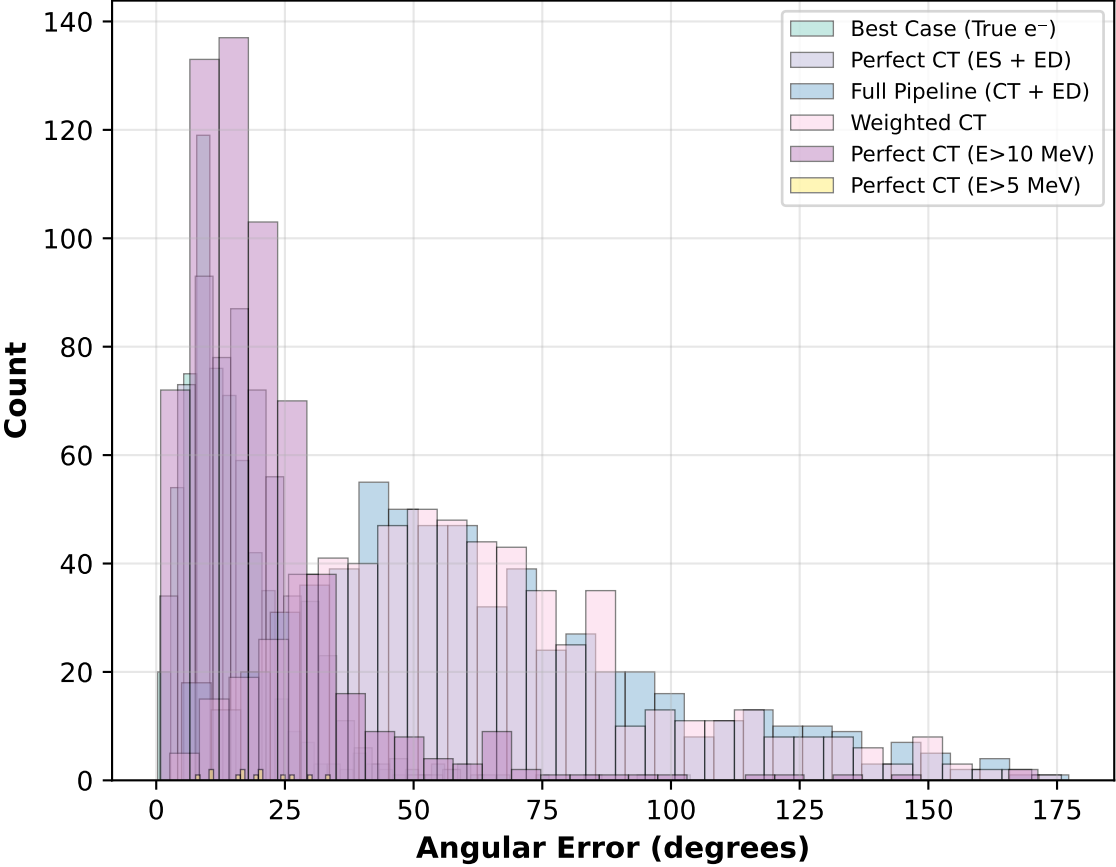
Summary Statistics

Scenario	N	Mean	Median	Std
Best Case (True e ⁻)	615	14.05°	11.61°	10.19°
Perfect CT (ES + ED)	615	17.37°	15.71°	11.35°
Full Pipeline (CT + ED)	618	63.15°	56.74°	34.46°
Weighted CT	618	62.99°	57.81°	33.34°
Perfect CT (E>10 MeV)	615	20.47°	16.38°	18.52°
Perfect CT (E>5 MeV)	13	19.51°	19.18°	7.41°

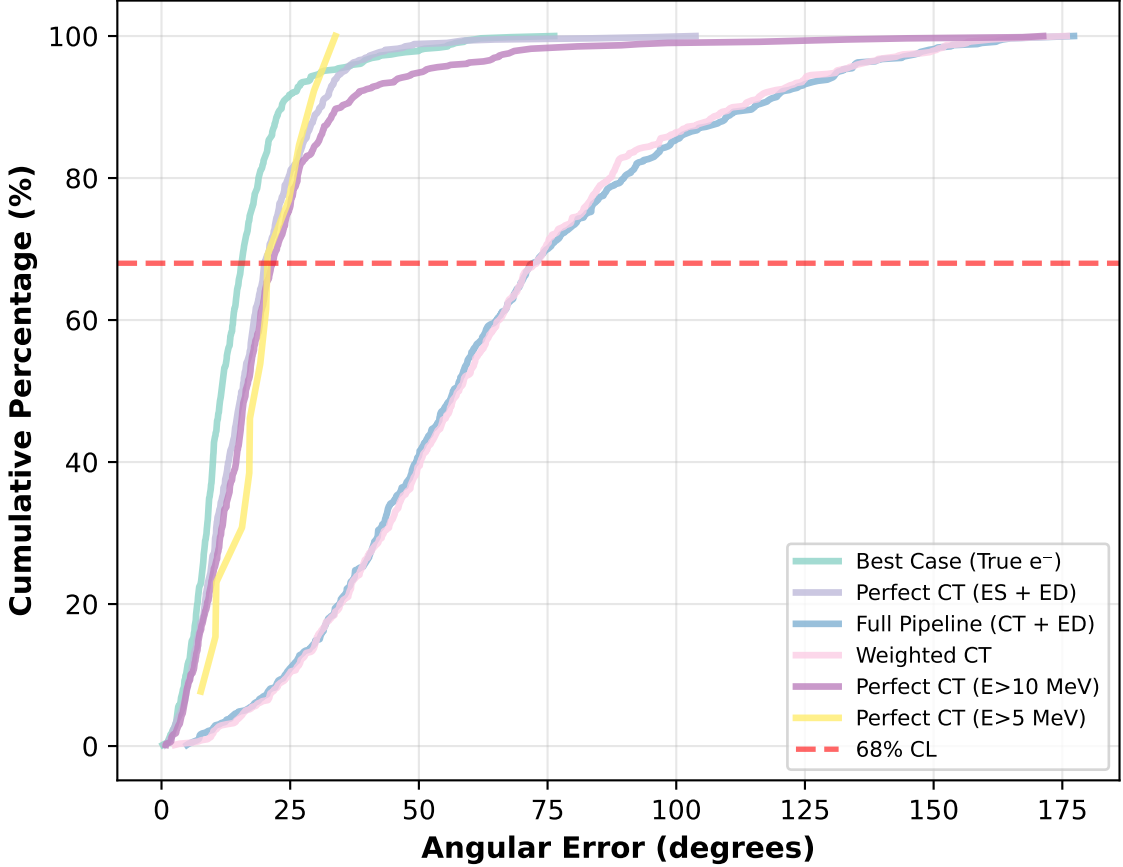
Mean Resolution ± Std Dev



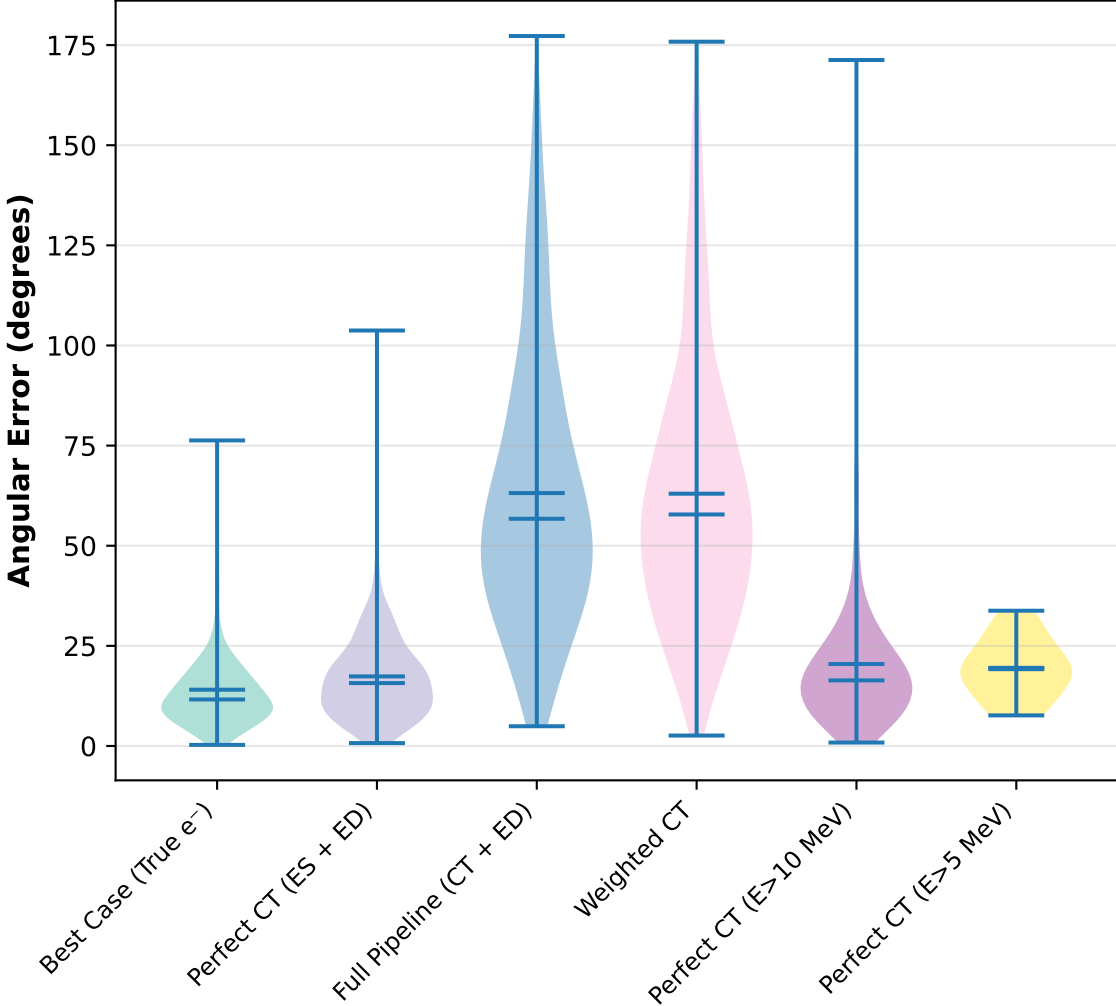
Distribution Overlay



Cumulative Distribution

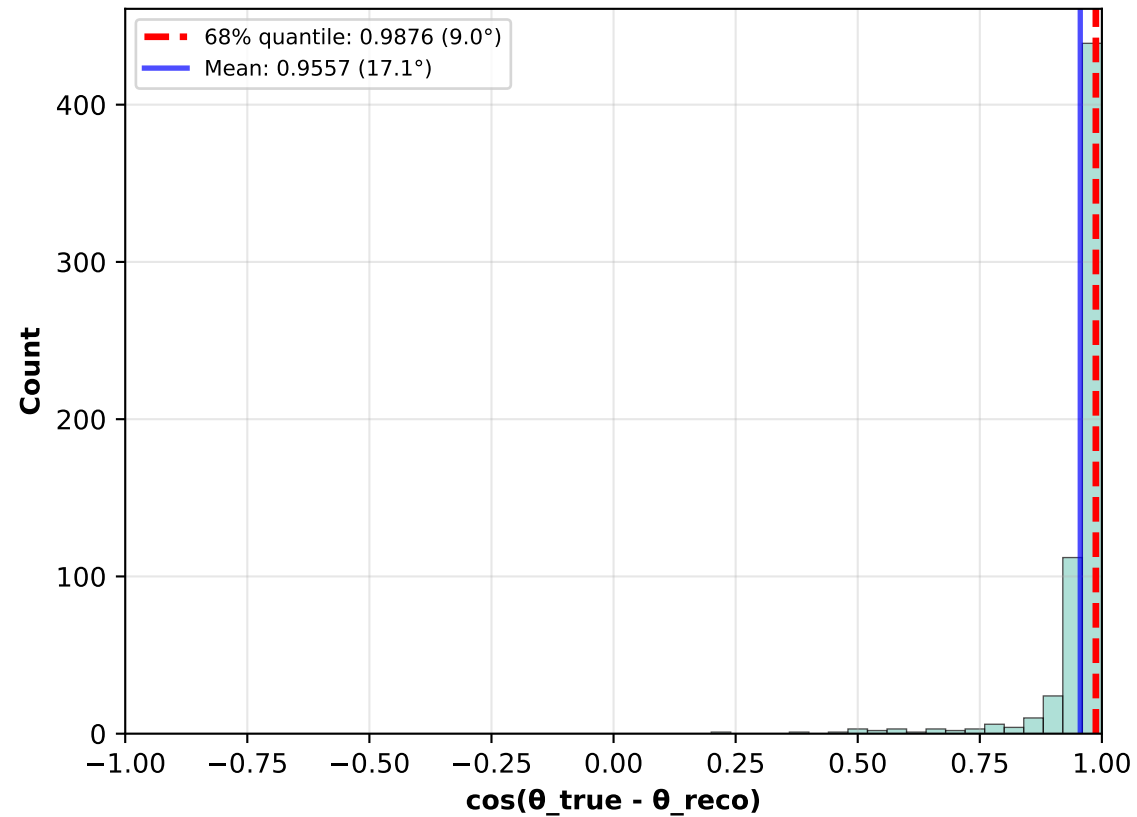


Distribution Shapes

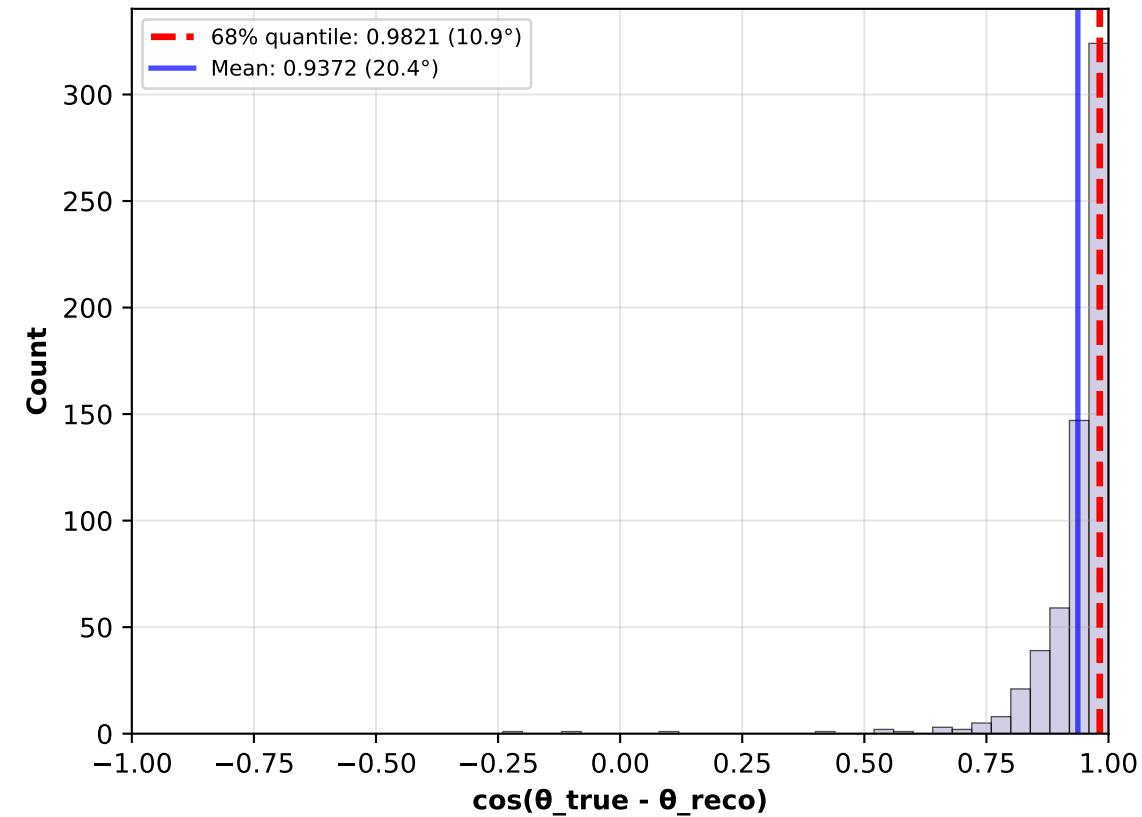


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Full Range

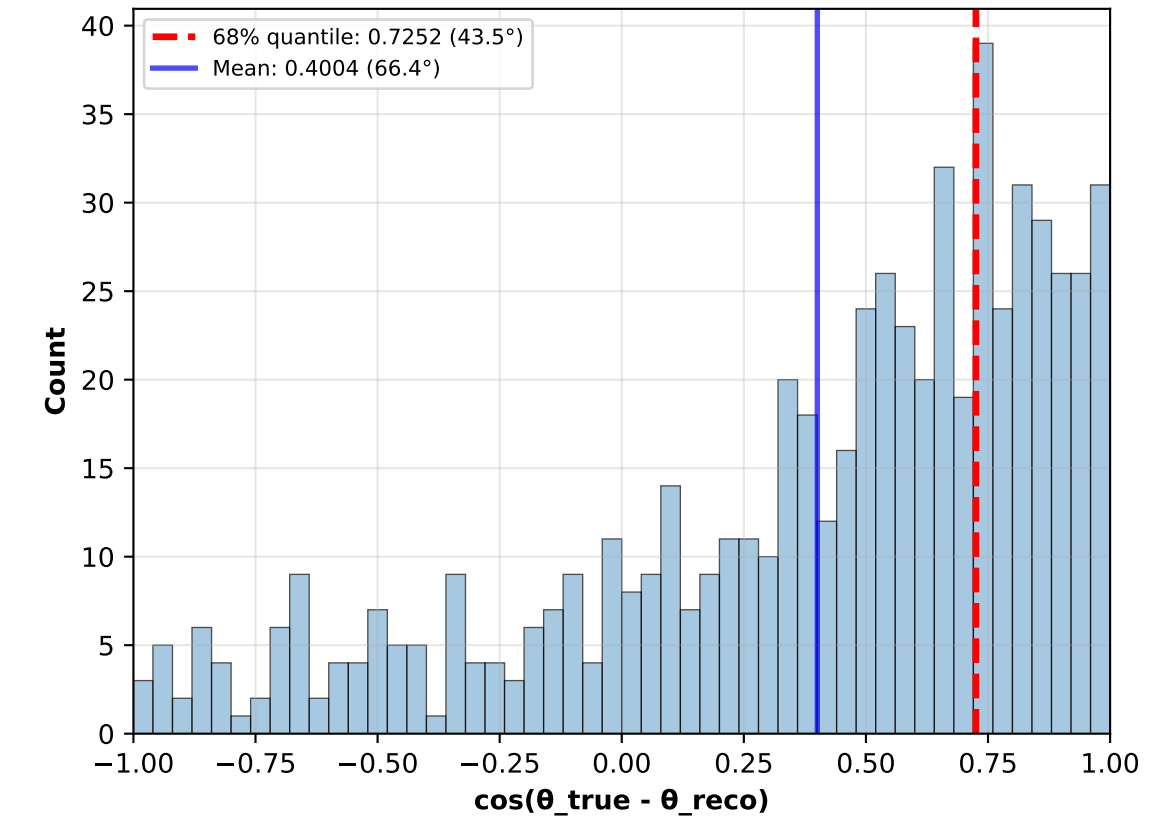
Best Case (True e^-)
(N=615)



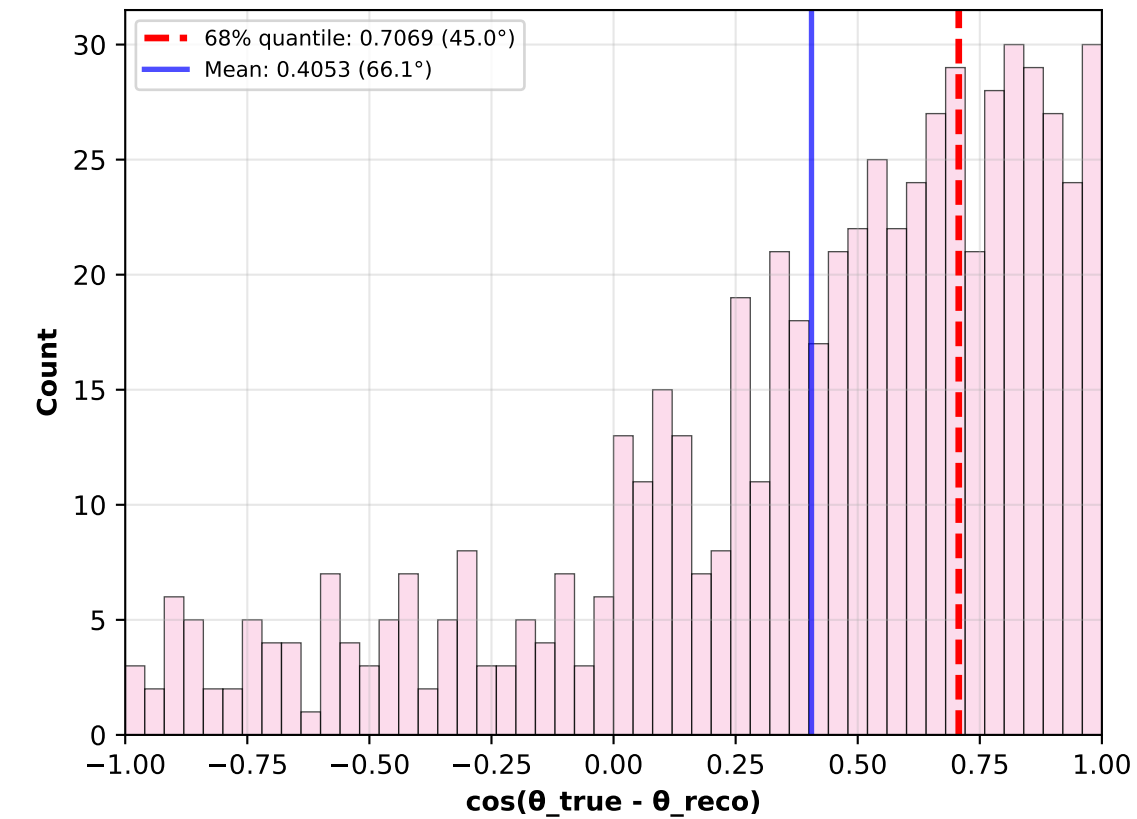
Perfect CT (ES + ED)
(N=615)



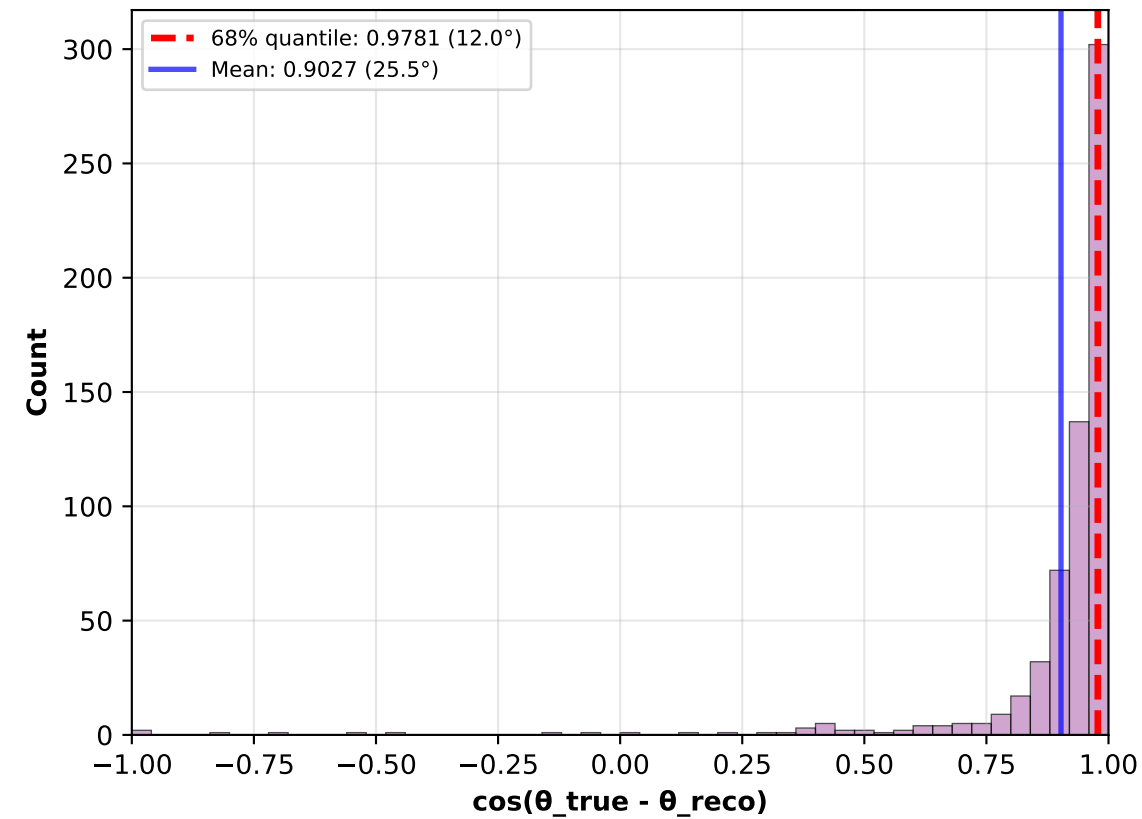
Full Pipeline (CT + ED)
(N=618)



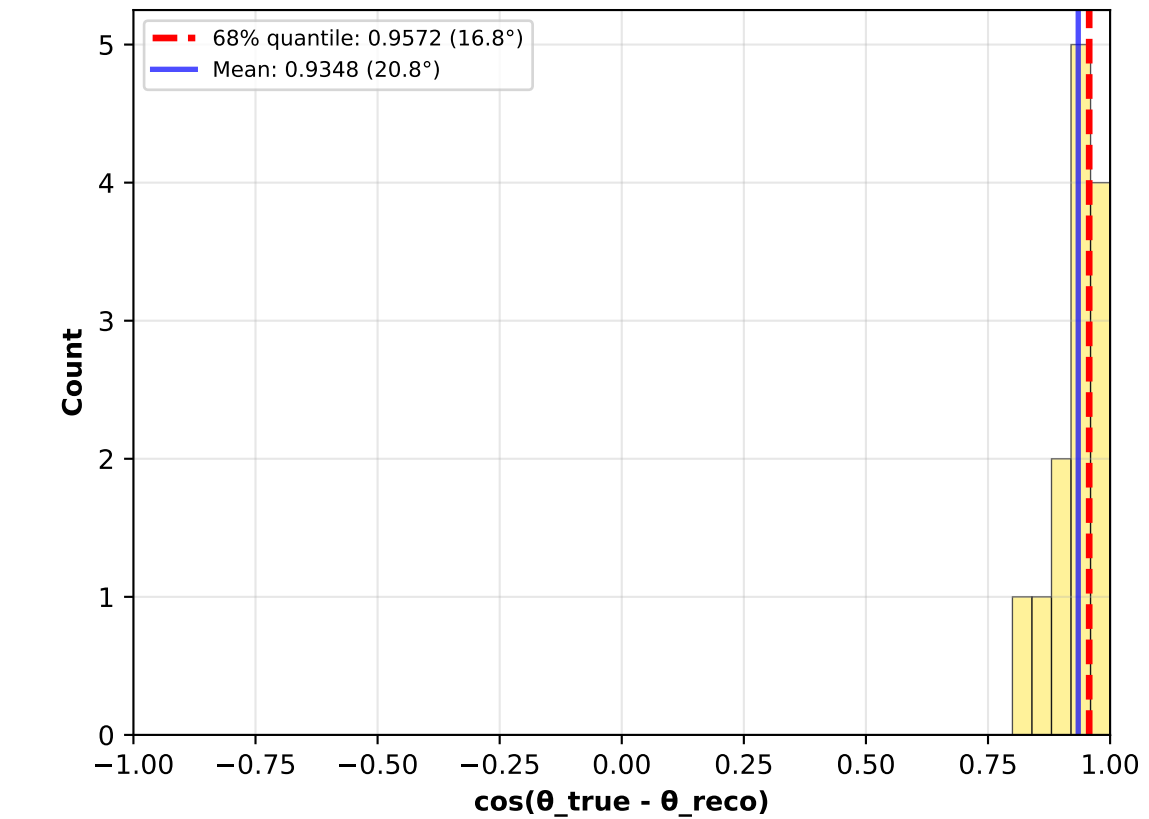
Weighted CT
(N=618)



Perfect CT (E>10 MeV)
(N=615)

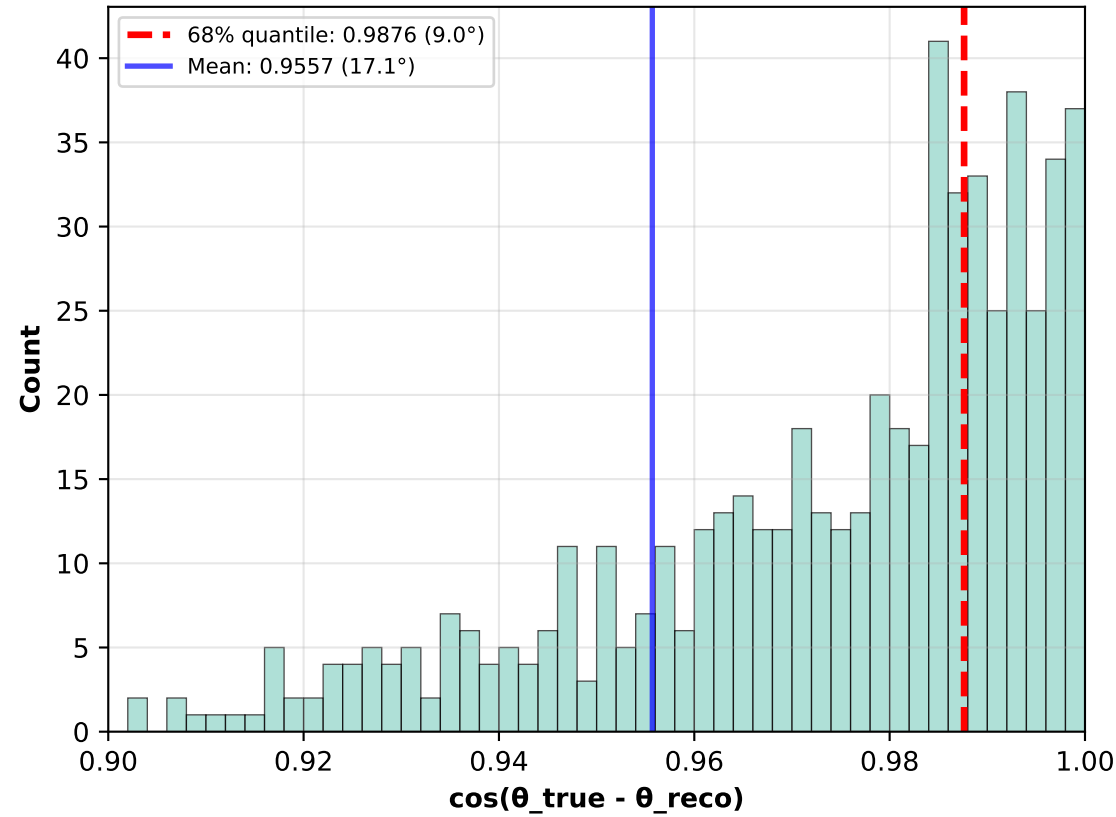


Perfect CT (E>5 MeV)
(N=13)

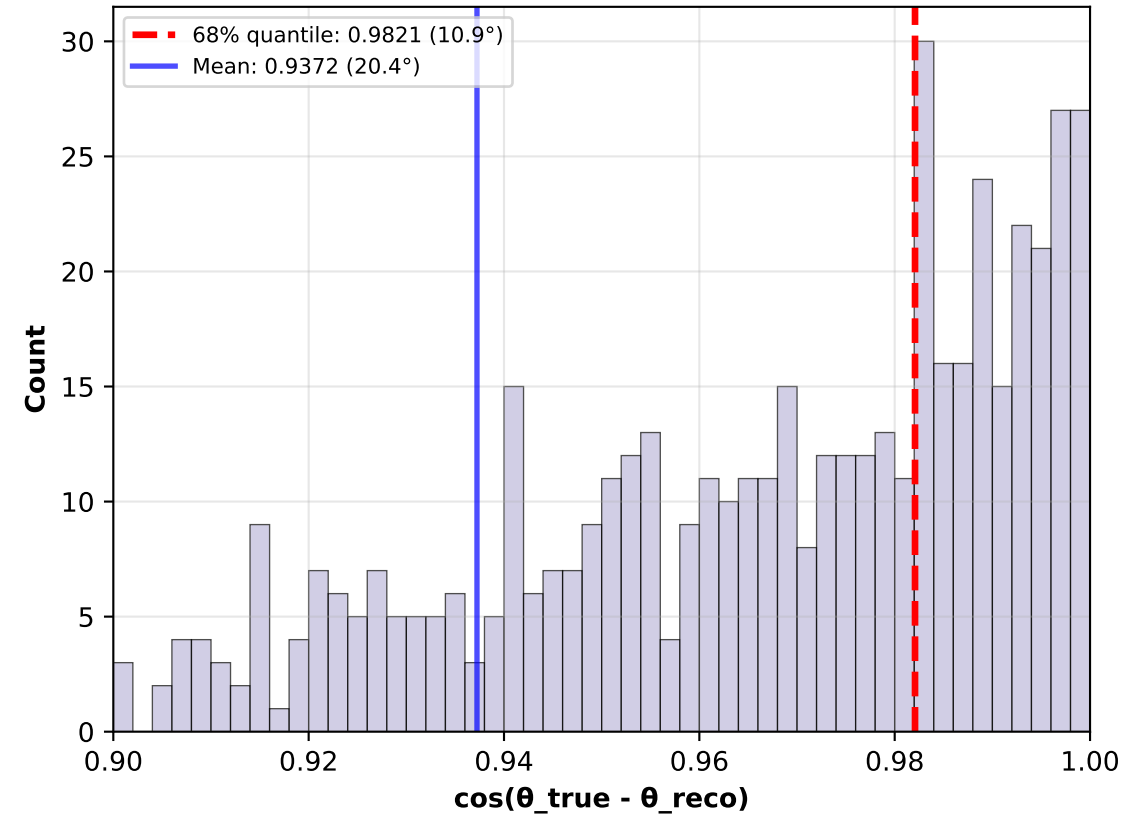


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Zoomed (0.9-1.0)

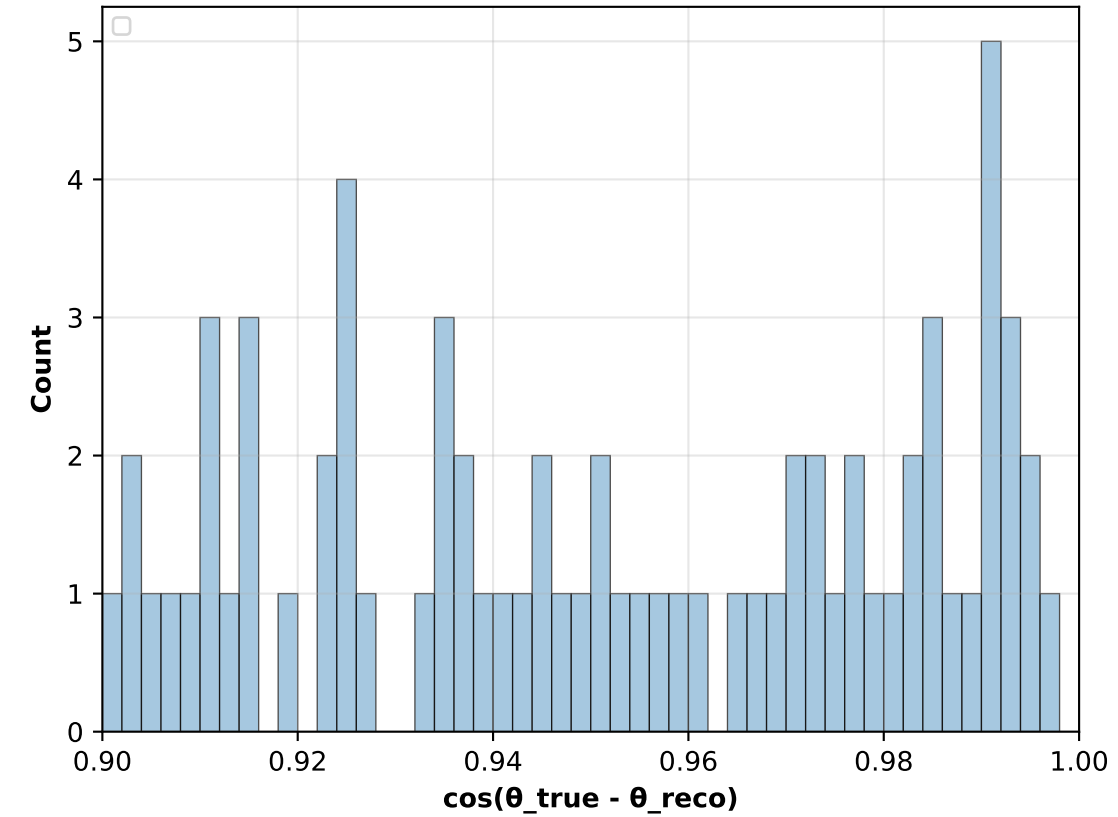
Best Case (True e^-)
(N=615)



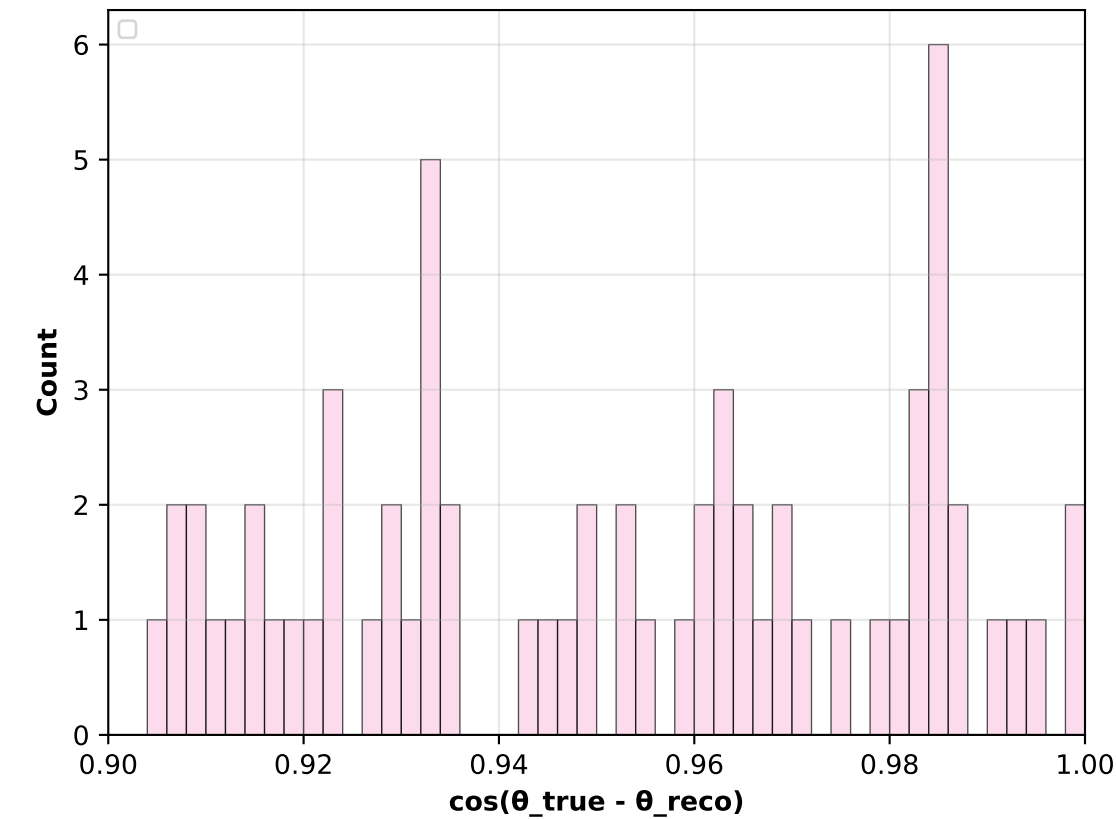
Perfect CT (ES + ED)
(N=615)



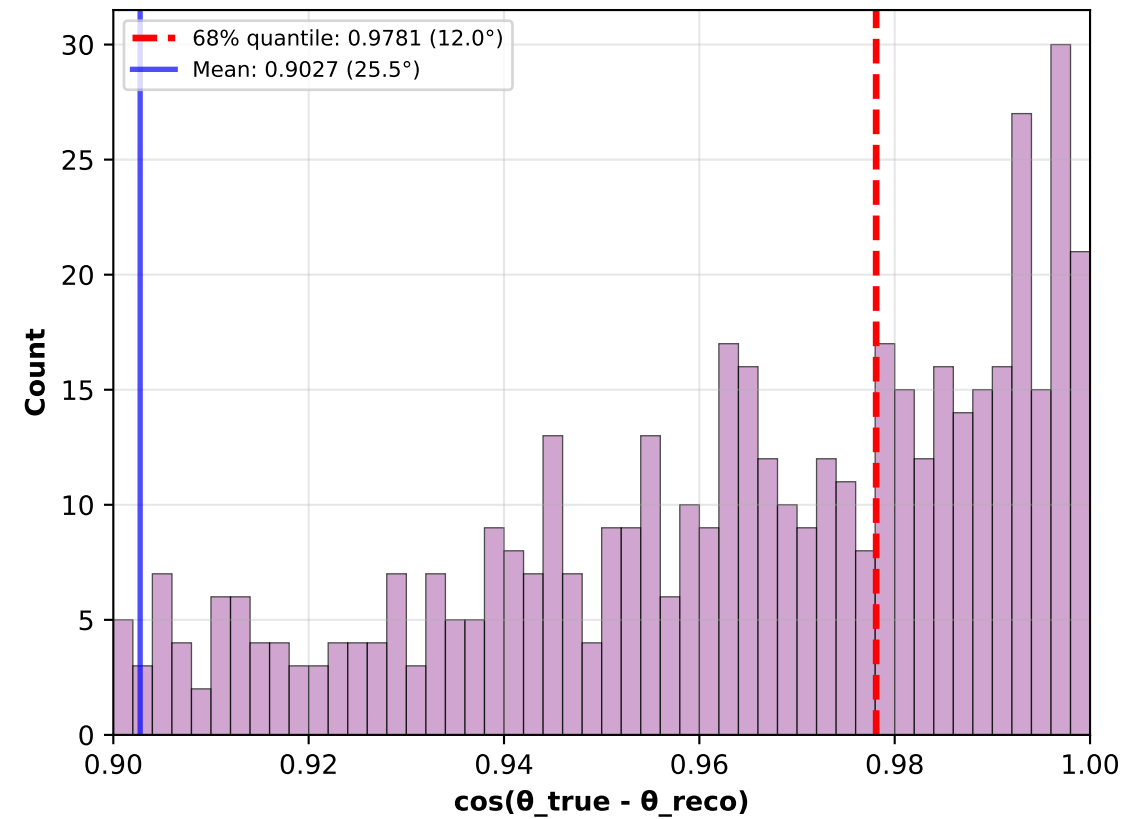
Full Pipeline (CT + ED)
(N=618)



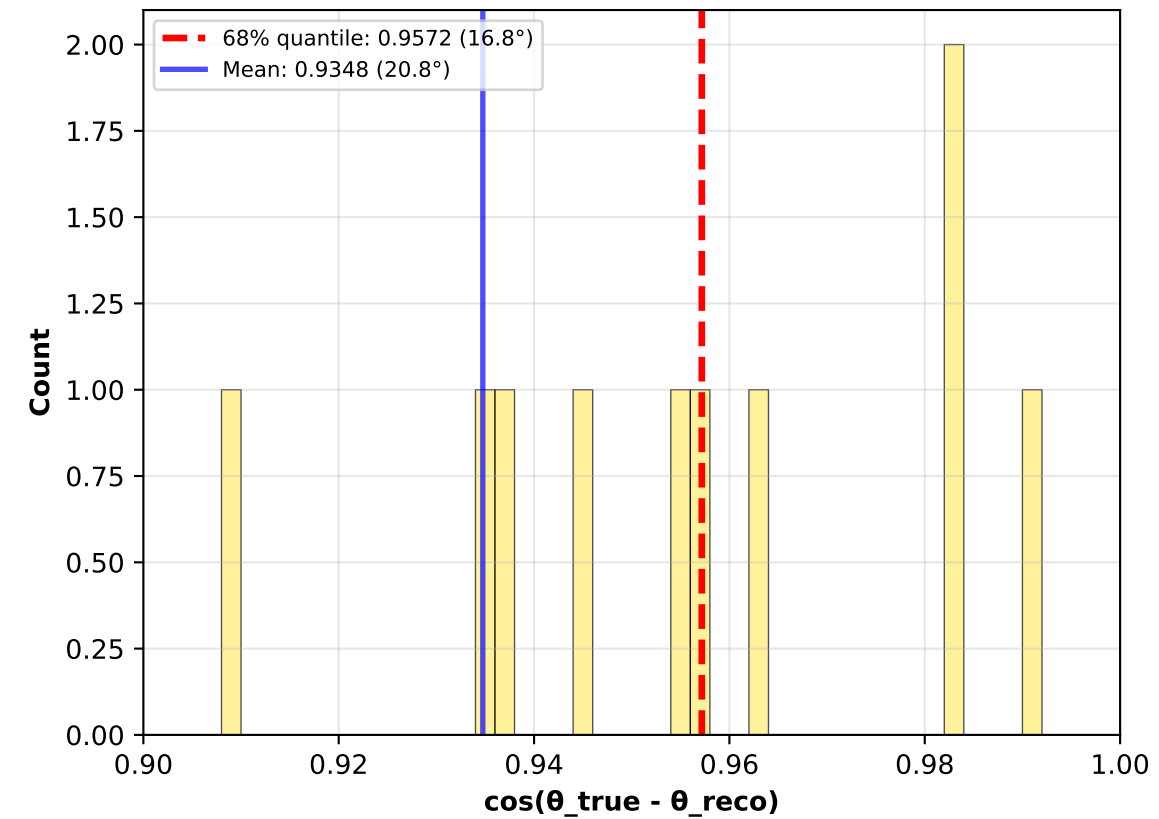
Weighted CT
(N=618)



Perfect CT (E>10 MeV)
(N=615)



Perfect CT (E>5 MeV)
(N=13)



Cluster Usage Across Scenarios

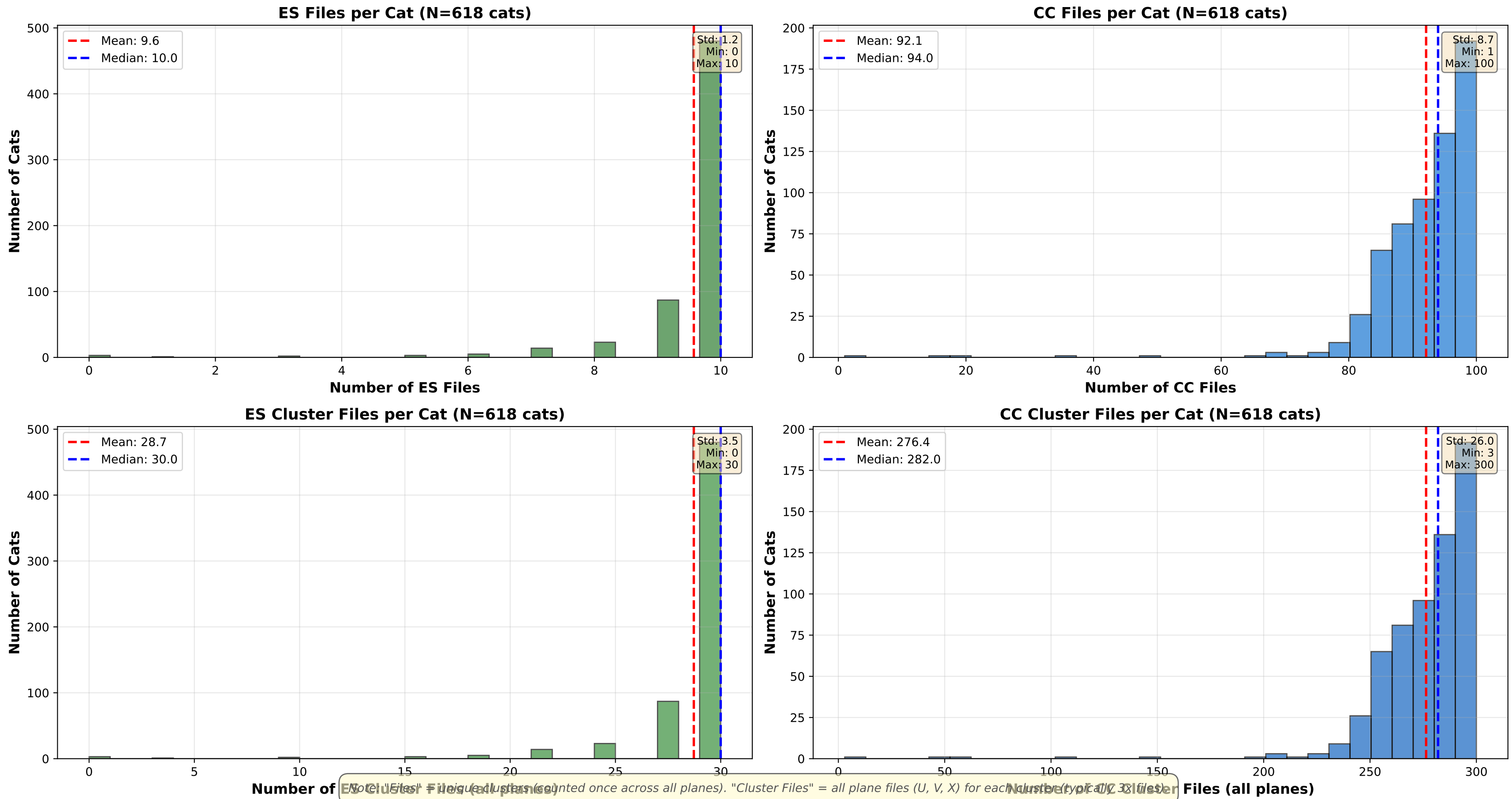
Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
Best Case (True e ⁻)	615	102	10	265	MCMC samples (weighted sum)
Perfect CT (ES + ED)	615	102	10	265	MCMC samples (weighted sum)
Full Pipeline (CT + ED)	618	102	10	3219	MCMC samples (weighted sum)
Weighted CT	618	102	10	3219	MCMC samples (weighted sum)
Perfect CT (E>10 MeV)	615	102	10	144	141.2% retention
Perfect CT (E>5 MeV)	13	102	10	236	MCMC samples (weighted sum)

Note: "Clusters Used" shows different meanings per scenario:

- Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
- Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
- Energy cut scenarios: Number of clusters passing energy threshold

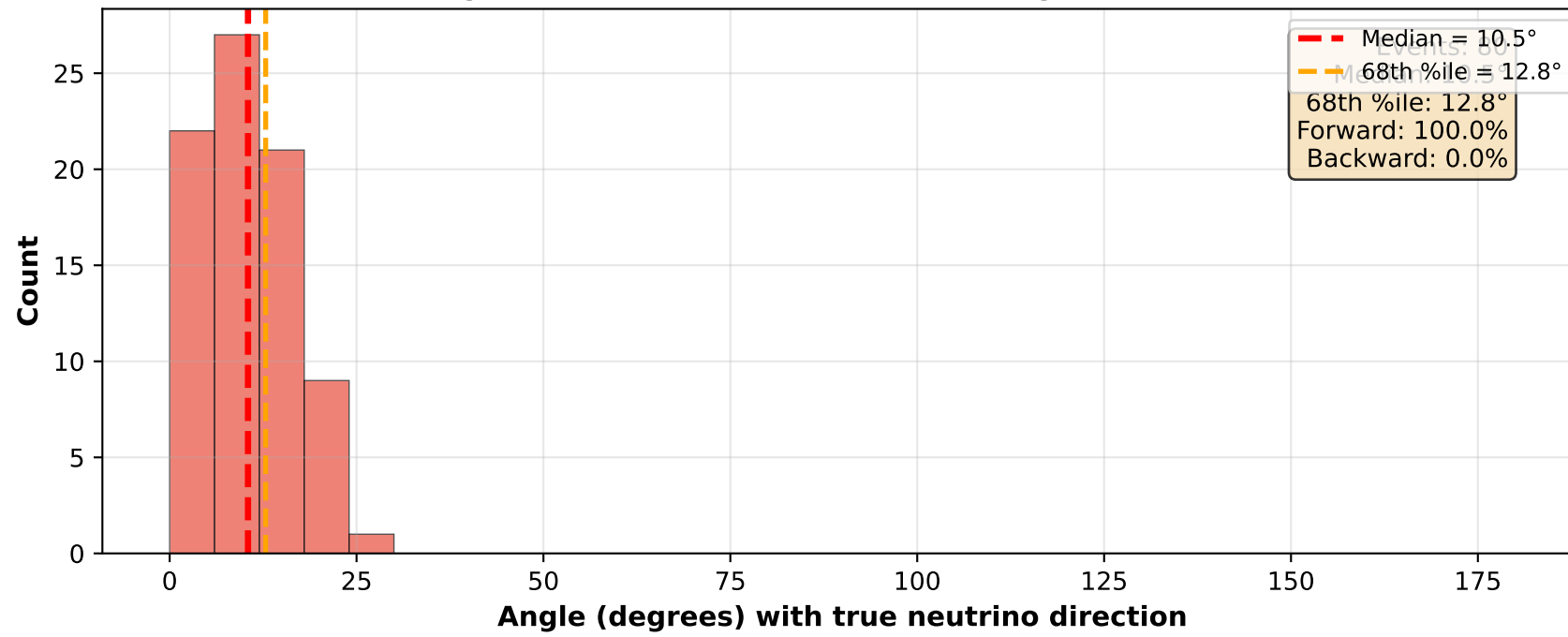
The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.

Input Data Statistics: Files and Clusters per Cat

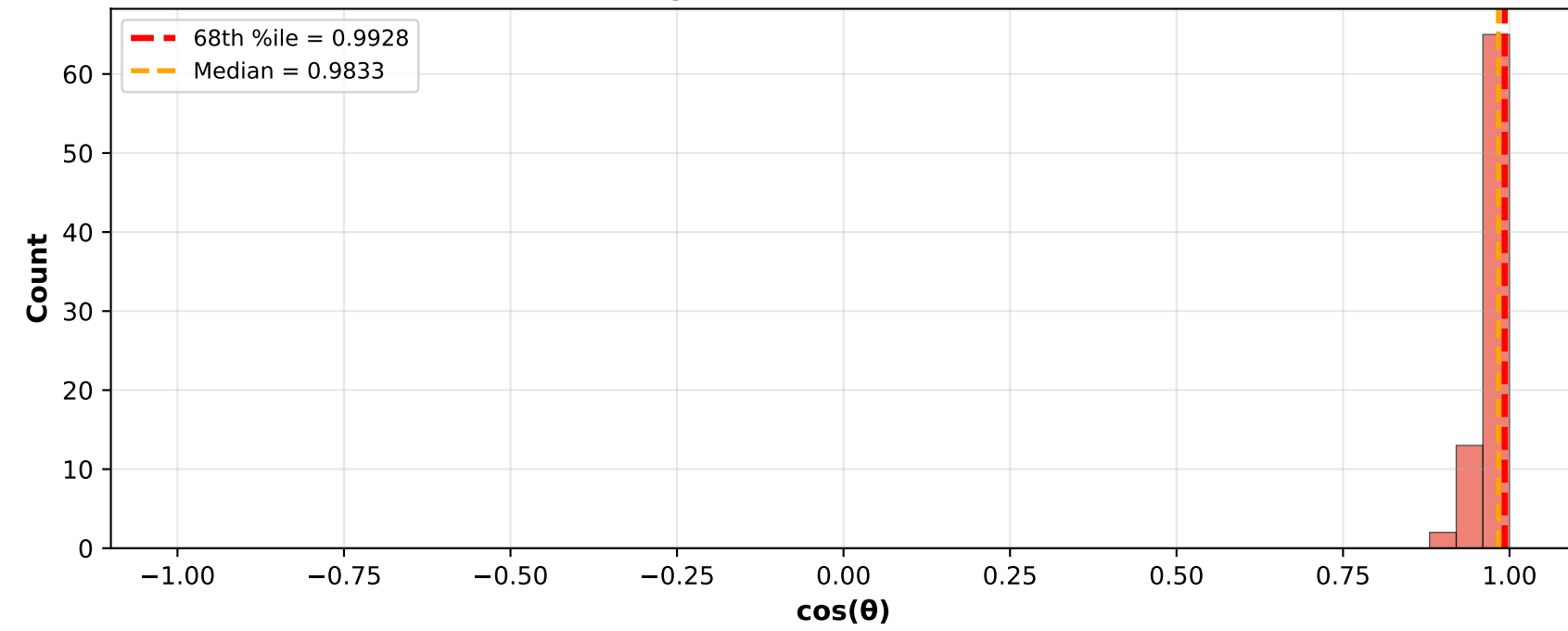


Perfect CT (ES + ED) - Axis-Aligned Analysis ($\pm 30.0^\circ$ selection)

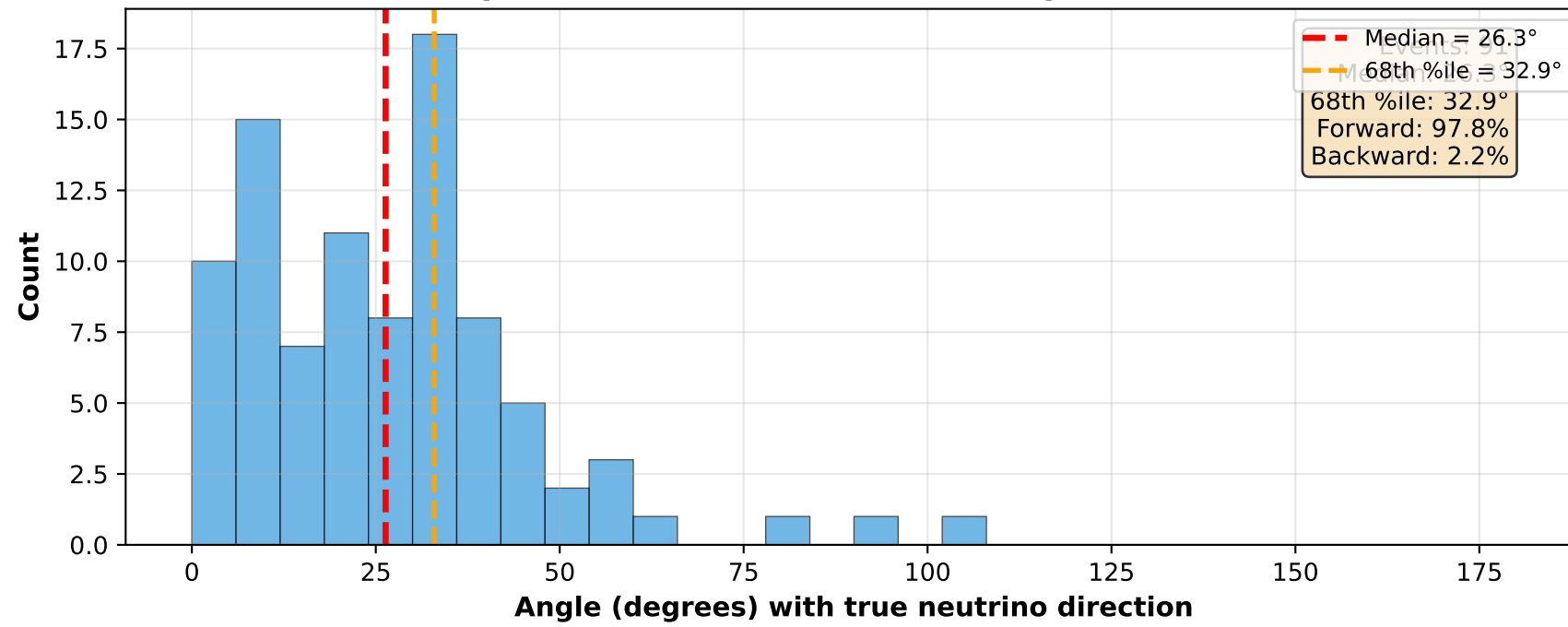
X-aligned (within $\pm 30.0^\circ$ of X-axis): Angle Distribution



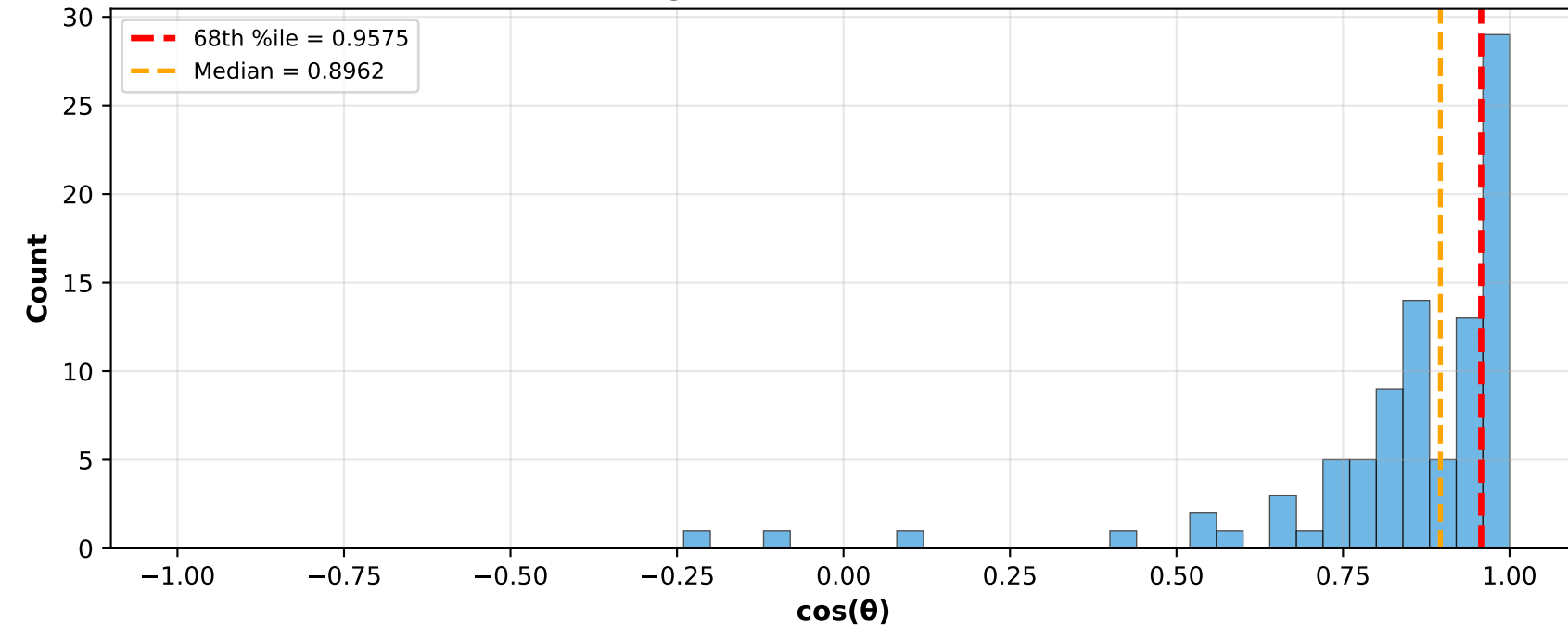
X-aligned: Cosine Distribution



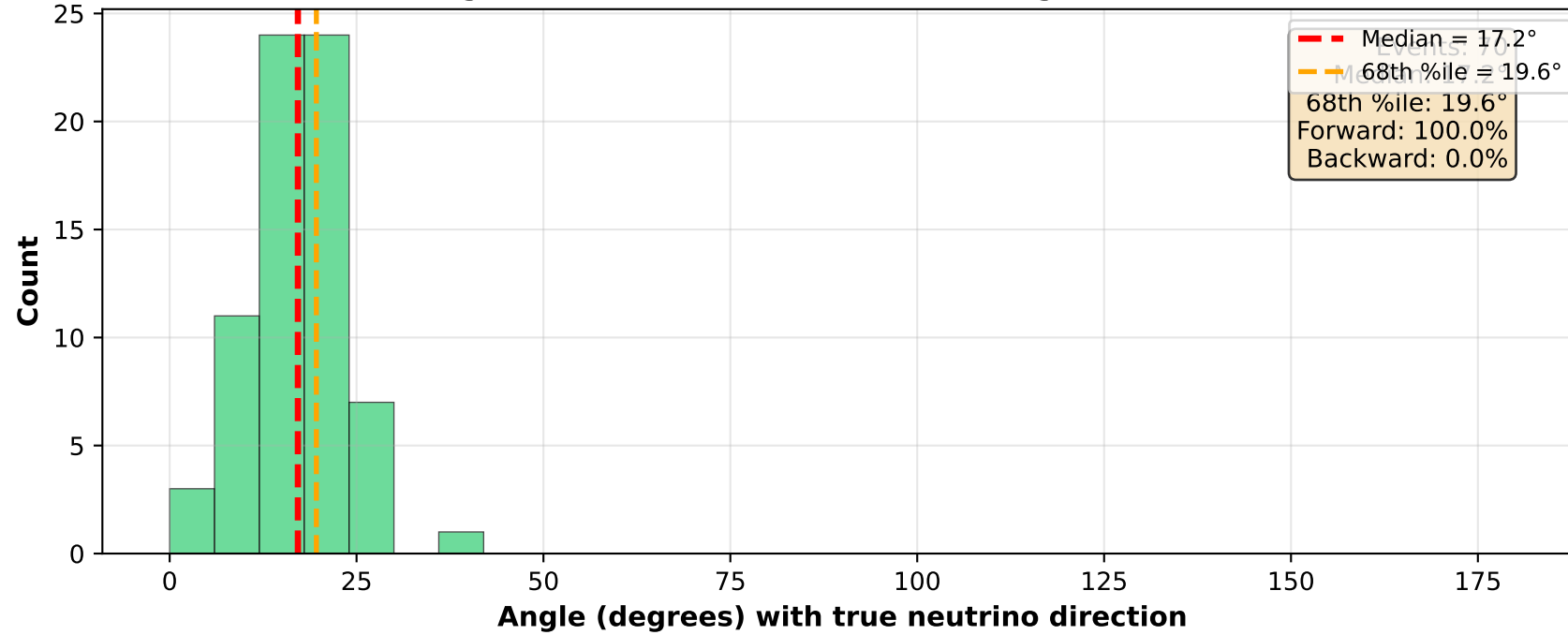
Y-aligned (within $\pm 30.0^\circ$ of Y-axis): Angle Distribution



Y-aligned: Cosine Distribution



Z-aligned (within $\pm 30.0^\circ$ of Z-axis): Angle Distribution



Z-aligned: Cosine Distribution

